Preprint

SUPPLEMENTARY MATERIAL

# The Energy Functional of the Constitutive Law $\mu(x) = x/(x+1)$

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**Abstract**

In the AQUAL field theory of Bekenstein and Milgrom the constitutive law $\mu$ enters through a kinetic function $F$, with $F'(y) = \mu(\sqrt{y})$ and $y = |\nabla\Phi|^2/a_0^2$. For $\mu(x) = x/(x+1)$ we show that $F$ is uniquely $F(y) = y - 2\sqrt{y} + 2\ln(1+\sqrt{y})$ once its irrelevant additive constant is fixed, and give its convergent expansions at both ends: Newtonian, $F = y - 2\sqrt{y} + \ln y + O(y^{-1/2})$, and deep-MOND, $F = \frac{2}{3}y^{3/2} - \frac{1}{2}y^2 + O(y^{5/2})$, where the leading term arises from a cancellation among all three terms. As a function of the field gradient, the energy density is twice continuously differentiable and strictly convex, with Hessian eigenvalues $2\mu$ and $2(x\mu)'$. The field equation is therefore elliptic wherever $\nabla\Phi \neq 0$, with the two eigenvalues within a factor of two of each other, and on a bounded domain with prescribed boundary values it has at most one solution.

## Setting

The AQUAL Lagrangian [1] is

$$\mathcal{L} = -\frac{a_0^2}{8\pi G}\, F\!\left(\frac{|\nabla\Phi|^2}{a_0^2}\right) - \rho\,\Phi. \tag{1}$$

Its Euler–Lagrange equation is

$$\begin{aligned} &\nabla\cdot\big[\mu(|\nabla\Phi|/a_0)\,\nabla\Phi\big] = 4\pi G\rho, \\ &\mu(\sqrt{y}) = F'(y). \end{aligned} \tag{2}$$

An additive constant in $F$ leaves Eq. (2) unchanged, and we fix it by $F(0) = 0$. Throughout,

$$\mu(x) = \frac{x}{x+1}. \tag{3}$$

## The Kinetic Function

**Theorem 1.** *The unique $F \in C^1[0,\infty)$ with $F'(y) = \mu(\sqrt{y})$ and $F(0) = 0$ is*

$$\boxed{F(y) = y - 2\sqrt{y} + 2\ln\big(1+\sqrt{y}\big).} \tag{4}$$

**Proof.** Two such functions differ by a constant, which $F(0) = 0$ removes. With $u = \sqrt{t}$,

$$\begin{aligned} F(y) &= \int_0^y \frac{\sqrt{t}}{1+\sqrt{t}}\,dt \\ &= 2\int_0^{\sqrt{y}} \Big(u - 1 + \frac{1}{1+u}\Big)\,du, \end{aligned} \tag{5}$$

since $u^2/(1+u) = u - 1 + 1/(1+u)$, and this integrates to Eq. (4). Conversely, with $u = \sqrt{y}$, $dF/du = 2u - 2 + 2/(1+u) = 2u^2/(1+u)$, so $F'(y) = (dF/du)/(2u) = u/(1+u)$, which extends continuously to $y = 0$. □

**Corollary 1 (the two limits).** *For $0 \le y \le 1$,*

$$\begin{aligned} F(y) &= 2\sum_{k=3}^{\infty} \frac{(-1)^{k+1}}{k}\, y^{k/2} \\ &= \tfrac{2}{3}\,y^{3/2} - \tfrac{1}{2}\,y^2 + \tfrac{2}{5}\,y^{5/2} - \cdots, \end{aligned} \tag{6}$$

*and for $y > 1$,*

$$F(y) = y - 2\sqrt{y} + \ln y + 2\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}\, y^{-k/2}. \tag{7}$$

**Proof.** Insert $\ln(1+u) = \sum_{k\ge1}(-1)^{k+1}u^k/k$ for $0 \le u \le 1$; the terms $k = 1, 2$ cancel $-2u$ and $u^2$. For $u > 1$ write $\ln(1+u) = \ln u + \ln(1+1/u)$ and expand the second logarithm. □

At high acceleration $F'(y) \to 1$ and Eq. (2) is Poisson's equation. At low acceleration the leading term $\frac{2}{3}y^{3/2}$ is the deep-MOND kinetic function [1], for which $\mu(x) \simeq x$ and the field equation becomes $\nabla\cdot(|\nabla\Phi|\nabla\Phi) = 4\pi G a_0\rho$. No single term of Eq. (4) produces it: the orders $y^{1/2}$ and $y$ cancel among all three.

## Convexity and Ellipticity

Write $q = \nabla\Phi/a_0 \in \mathbb{R}^3$ and $W(q) = F(|q|^2)$, so that the energy density is $(a_0^2/8\pi G)\,W(q) + \rho\Phi$ and Eq. (2) reads $\nabla\cdot A(\nabla\Phi) = 4\pi G\rho$ with $A(p) = \mu(|p|/a_0)\,p = \frac{a_0}{2}\nabla W(p/a_0)$.

**Theorem 2.** *$W \in C^2(\mathbb{R}^3)$, smooth away from $q = 0$. Its Hessian at $q \neq 0$, with $x = |q|$ and $\hat{q} = q/x$, is*

$$\nabla^2 W = 2\mu(x)\,\big(I - \hat{q}\hat{q}^{\,T}\big) + 2\,(x\mu)'(x)\,\hat{q}\hat{q}^{\,T}, \tag{8}$$

*with $2\mu(x) = 2x/(1+x)$ twice and $2(x\mu)'(x) = 2x(x+2)/(x+1)^2$ once, both positive, and $\nabla^2 W(0) = 0$. Hence $W$ is strictly convex, $A$ is strictly monotone, $(A(p) - A(p'))\cdot(p - p') > 0$ for $p \neq p'$, and the field equation is elliptic wherever $\nabla\Phi \neq 0$, with*

$$1 < \frac{(x\mu)'}{\mu} = \frac{x+2}{x+1} \le 2. \tag{9}$$

**Proof.** $W(q) = g(|q|)$ with $g(r) = F(r^2) = r^2 - 2r + 2\ln(1+r)$, so $g'(r) = 2r^2/(1+r) = 2r\mu(r)$ and $g''(r) = 2(r\mu)'(r) = 2r(r+2)/(1+r)^2$. For a radial function, $\nabla W = g'(r)\,\hat{q}$ and $\nabla^2 W = (g'/r)(I - \hat{q}\hat{q}^{\,T}) + g''\,\hat{q}\hat{q}^{\,T}$, which is Eq. (8). As $r \to 0$, $g'/r = 2\mu(r) \to 0$ and $g'' \to 0$, so $\nabla W$ and $\nabla^2 W$ extend continuously to $q = 0$ with value 0, and $W \in C^2$; it is smooth for $q \neq 0$. For $p \neq p'$ let $\varphi(t) = W(p' + t(p - p'))$. Then $\varphi''(t) = (p-p')^T\,\nabla^2 W\,(p-p') > 0$ except where $p' + t(p-p') = 0$, which happens for at most one $t$, so $W$ is strictly convex and $\varphi'(1) - \varphi'(0) = \int_0^1 \varphi'' > 0$, which is the monotonicity of $\nabla W$ and hence of $A$. The eigenvalues of Eq. (8) are those of the principal symbol of $\nabla \cdot A(\nabla\Phi)$, up to the factor 2, and Eq. (9) follows from Eq. (3). □

At $\nabla\Phi = 0$ both eigenvalues vanish: the equation is degenerate there, as the deep-MOND form $\nabla \cdot (|\nabla\Phi|\nabla\Phi)$ already is.

**Corollary 2 (uniqueness).** *Let $\Omega \subset \mathbb{R}^3$ be a bounded connected domain with piecewise smooth boundary. Two solutions $\Phi_1, \Phi_2 \in C^2(\Omega) \cap C^1(\bar{\Omega})$ of Eq. (2) with $\Phi_1 = \Phi_2$ on $\partial\Omega$ coincide.*

**Proof.** $A$ is $C^1$, including at $p = 0$, where $A(p) = |p|\,p/(a_0 + |p|)$ has derivative 0. Let $w = \Phi_1 - \Phi_2$, which vanishes on $\partial\Omega$. By the divergence theorem and Eq. (2),

$$\int_\Omega \big(A(\nabla\Phi_1) - A(\nabla\Phi_2)\big) \cdot \nabla w \, d^3x = 0. \qquad (10)$$

The integrand is continuous and, by Theorem 2, positive wherever $\nabla\Phi_1 \neq \nabla\Phi_2$. Hence $\nabla w = 0$ on $\Omega$, $w$ is constant, and $w = 0$. □

The same argument applies on all of space whenever the boundary term at infinity vanishes; we do not treat that case here.

**Summary**

The constitutive law $\mu(x) = x/(x+1)$ has the closed-form kinetic function $F(y) = y - 2\sqrt{y} + 2\ln(1+\sqrt{y})$, unique up to the constant that the field equation cannot see. It reduces to Poisson's equation at high acceleration and to the deep-MOND equation at low acceleration, through convergent expansions given in full. As a function of the field gradient the energy is $C^2$ and strictly convex, so the field equation is elliptic away from $\nabla\Phi = 0$ and its solutions are unique for given boundary values.

**References**

[1] J. Bekenstein and M. Milgrom, *Does the missing mass problem signal the breakdown of Newtonian gravity?*, Astrophys. J. **286** (1984) 7.